— *Solution notes* —

COMPUTER SCIENCE TRIPOS Part IB – 2024 – Paper 5

3 Computer Networking (awm22)

Introduction topic and specifics given in Data-Link topic

(*a*) Define the terms latency and capacity as applied to communication channels [2 marks]

Answer: Latency propagation time for data to travel along the link (seconds)

Bandwidth number of bits sent (or received) per unit time (bits/sec or bps)

(*b*) Is there a strict relation between the two? [1 mark]

Answer: No.

(*c*) Imagine a packet-based network channel, with fixed bandwidth and fixed latency, providing service to a transport protocol.

(*i*) In this context, describe a basic system that provides reliable transmission of packet data. Include a description of the essential features of your approach. [8 marks]

Answer:

I'm asking for a simple ARQ to be described here.

Such a mechanic for reliable delivery will imply
checking the correctness (and discarding upon corruption)(2)
ensuring the reliable delivery (acknowledgement of delivery)(2)
time-out to identify events of lost packets or lost ACKs(2)
a single bit window is required to ensure duplicate and missing packets/acks can be identified(2)

(*ii*) The network channel will impact the effective throughput of your reliable transport protocol. Outline an example of why this occurs. [2 marks]

Answer: An important part here is "reliable transport protocol" - reminding students we have an acknowledgement scheme in play.

the classic reliable delivery protocol taught in the course is "ARQ" which utilises explicit acknowledgement of receipt of uncorrupted data.

the data and ack will require a complete round trip - thus to achieve reliability in transport the latency impact is made twice for each packet+ack

(2 marks)

(*iii*) Describe a straightforward extension to your protocol that would improve the performance of your transport protocol without changing the channel provided by the underlying network. [4 marks]

Answer:

For ARQ the anticipated answer is "using a windowing mechanic" (2 marks) - thereby allowing more than one packet to be outstanding at any time - this reduces the overt latency experienced in the channel by hiding it utilising packets-in-flight. (2 marks)

(*iv*) In what circumstances might these improvements have only limited benefit? [3 marks]

Answer:

Very short links, or, in general, channels with low bandwidth-delay products (1 marks)

Channels with significant packet loss rates. (1 mark)

Use cases where there is insufficient data outstanding (1 mark)